

1) If $(1, \dots, 1) \in C$,

$$2^{n-k} \leq V(n, R-1) + \frac{n+2 - \left\lfloor \frac{d}{2} \right\rfloor}{n+R+4 - \left\lfloor \frac{d}{2} \right\rfloor} \binom{n}{R}. \quad (4.13)$$

2) If $(1, \dots, 1) \notin C$,

$$2^{n-k} \leq V(n, R-1) + \frac{\left(n - \frac{2}{3}(R-2) - \left\lfloor \frac{d}{2} \right\rfloor\right) \binom{n}{R} + \frac{1}{3}(R+1) \binom{n-1}{R-1}}{n+3 - \left\lfloor \frac{d}{2} \right\rfloor}. \quad (4.14)$$

Proof: By (2.8),

$$\begin{aligned} & \left(n+1 - R - \left\lfloor \frac{d}{2} \right\rfloor\right) (V(n, R) - 2^{n-k}) \\ & \geq \sum_{s \in A_1^*} [(R+1)|T^*(s)| - (n+1)] \\ & \quad + \sum_{s \in A_2^*} [(R+1)|T^*(s)| - (n+1)] \\ & \geq |A_1^*| + (R+2)|A_2^*| \\ & = (R+2)|A_1^*| - (R+1)|A_2^*| \\ & \geq (R+2)[2^{n-k} - V(n, R-1)] - (R+1)|A_2^*|. \end{aligned}$$

Then apply Lemma 2. $\square$

Equation (4.13) is stronger than (4.14) since

RHS of (4.14) - RHS of (4.13)

$$\begin{aligned} & \geq \frac{\frac{1}{3}(R+1) \binom{n}{R}}{n+3 - \left\lfloor \frac{d}{2} \right\rfloor} - \binom{n}{R} \frac{(R+1)(R+2)}{\left(n+3 - \left\lfloor \frac{d}{2} \right\rfloor\right) \left(n+R+4 - \left\lfloor \frac{d}{2} \right\rfloor\right)} \\ & = \frac{(R+1)}{n+3 - \left\lfloor \frac{d}{2} \right\rfloor} \binom{n}{R} \left[\frac{1}{3} - \frac{R+2}{n+R+4 - \left\lfloor \frac{d}{2} \right\rfloor} \right] \geq 0. \end{aligned}$$

Corollary 3: Let C be an $[n, k]R$ code with minimal distance $d > 2$, and $n+1 = t(R+1) - 1$ ($t > 2$ even). Then we always have

$$2^{n-k} \leq V(n, R-1) + \frac{\left(n - \frac{2}{3}(R-2) - \left\lfloor \frac{d}{2} \right\rfloor\right) \binom{n}{R} + \frac{1}{3}(R+1) \binom{n-1}{R-1}}{n+3 - \left\lfloor \frac{d}{2} \right\rfloor}. \quad (4.14)$$

As noticed in Remark 1) following Theorem 2, for any optimal $[n, k]R$ code with $R > 1$, we may assume its minimal distance $d > 2$. Hence the above results apply. Equation (4.14) eliminates the existence of $[40, 18]_6$, $[46, 12]_{11}$, $[46, 36]_2$, $[62, 33]_7$ optimal codes. This gives the following new results on the table of $t[n, k]$ ($n \leq 64$) in [5].

Theorem 4: $t[40, 18] \geq 7$, $t[46, 12] \geq 12$, $t[46, 36] = 3$, $t[62, 33] \geq 8$.

As for the table of $l(m, R)$ ($R \leq m \leq 24$) in [1], we have the following.

Corollary 4: $l(22, 6) \geq 41$, $l(10, 2) \geq 47$.

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Constructions of Error-Correcting DC-Free Block Codes

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Abstract—A $(2n, l, c, d)$ dc-free binary block code is a code of length $2n$, constant weight n , maximum runlength of a symbol l , maximum accumulated charge c , and minimum distance d . The requirements are that l and c will be small. We present two dc-free codes with distance $2d$, $b \geq 1$ length $2n + 2r(d-1)$ for $d \leq 3$ and length $2n + 2r(d-1)(2d-1)$ for $d > 3$, where $r \leq \lceil \log_2(2n+1) \rceil$. For the first code $l=4$, $c=2$, and the asymptotic rate of this code is 0.7925. For the second code $l=6$, $c=3$, and the asymptotic rate of this code is 0.8858. Asymptotically, these rates achieve the channel capacity. For small values of n these codes does not achieve the best rate. As an example of codes of short length with good rate, we first present a $(30, 10, 6, 4)$ dc-free block code with 2^{21} codewords. Finally, we present a construction for which from a given code C_1 of length n , even weight, and distance 4, we obtain a $(4n, l, c, 4)$ dc-free block code C_2 , where l is 4, 5 or 6, and c is not greater than $n+1$ (but usually significantly smaller). The codes obtained by this method have good rates for small lengths. We discuss the encoding and decoding procedures for all the codes.

I. INTRODUCTION AND DEFINITIONS

(n, d) code is a binary code of length n and distance d . An (n, d, w) code is an (n, d) code in which the codewords have constant weight w . A dc-free block code with parameters $(2n, l, c, d)$ is a $(2n, d, n)$ code, where the maximum length of a string of consecutive 0's or 1's in any sequence of codewords is l , and the maximum absolute value of the difference between the number of 0's and 1's in any prefix of any codeword, called the

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accumulated charge, is c . A $(2n, k, l, c, d)$ code is a $(2n, l, c, d)$ code with 2^k codewords. These parameters determine an input restricted channel and these codes are used in magnetic recording channels to achieve dc freeness together with error correction. The goal is to have a large rate of $k/2n$, large d , together with small l and c [1]. Lower bounds for constant weight codes were given by various authors, e.g., [2]–[8]. But usually, the codes that are constructed do not seem to have easy constructive methods or easy encoding and decoding procedures. Also, we do not know the parameters l and c of these codes.

Ferreira [9] gave a construction for a $(16, 8, 8, 5, 4)$ dc-free block code. Blaum [10] gave constructions for a $(16, 9, 6, 5, 4)$ dc-free block code and a $(16, 9, 8, 4, 4)$ dc-free block code. He also gave easy procedures for encoding and decoding. Calderbank, Herro, and Telang [11], and van Tilborg and Blaum [12], gave some constructions that achieve better rates, and also can be applied for distances greater than 4. Their constructions involve concatenations of a few blocks of length $2r$ and weight r . We also make our codewords as concatenation of a few blocks. But, the blocks might have different lengths and different weights. By using these kinds of blocks we make a significant improvement of the rate.

Throughout this correspondence we consider each codeword as a concatenation of k blocks $[B_1 B_2 \cdots B_k]$. We say that a word has a $(w_1, w_2, \dots, w_k)$ configuration if B_i has weight w_i . Each codeword of our codes will have two parts. The I -part that is the information part and the C -part that is the checkbits part. The C -part is computed from the I -part. By choosing appropriate checkbits we will construct codes with desired distances and with relatively easy decoding procedures. In Section II we present some methods to compute checkbits for the codes. For each method we will give the procedure for error detection and correction.

In Section III we construct two dc-free codes with distance $2d$, $d \geq 1$, length $2n + 2r(d-1)$ for $d \leq 3$ and length $2n + 2r(d-1)(2d-1)$ for $d > 3$, where $r \leq \lceil \log(2n+1) \rceil$ (through this correspondence all logarithms are taken to the base 2). The first code has maximum runlength 4, maximum accumulated charge 2, and asymptotic rate of 0.7925. The second code has maximum runlength 6, maximum accumulated charge 3, and asymptotic rate of 0.8858. These rates are significantly higher from the ones obtained by Calderbank, Herro, and Telang [11], and by van Tilborg and Blaum [12]. In fact the rates of these codes achieve the channel capacity asymptotically as the code length goes to infinity. We also show that the encoding and decoding procedures for these codes are relatively easy. For small values of n the rates of these codes are not the best and other construction techniques can be used to achieve good rates. In Section IV we generate a $(30, 21, 10, 6, 4)$ dc-free code. In Section V we obtain a dc-free block code of length $4n$, maximum runlength 4, 5, or 6, maximum accumulated charge not greater than $n+1$ (but usually significantly smaller), and distance 4. This code is constructed from an (n, d) code C . We do not know the cardinality of the codes, but for all the cases $16 \leq 4n \leq 32$ and $4n = 64$ they are not far from the known lower bounds. If C has a simple construction then the dc-free code obtained from it will have relatively easy procedures for encoding and decoding. For example, by this method a $(16, 9, 4, 2, 4)$ dc-free block code is obtained.

In the sequel we also need the following definitions. Let $A(n, d)$ denote the maximum number of codewords in an (n, d) code. Similarly we define $A(n, d, w)$ and $A(2n, l, c, d)$. For two binary words, X and Y , of the same length, let $d(X, Y)$ denote

the Hamming distance between X and Y , and $W(X)$ denote the weight of X . For a received codeword $B = b_0 \cdots b_{n-1}$ we say that a 1-error occurred in b_s if $b_s = 1$ but in the transmitted codeword bit s was 0. We say that a 0-error occurred in b_s if $b_s = 0$ but in the transmitted codeword bit s was 1.

II. THE CHECKBITS

In this section we describe some methods to compute checkbits to construct a code with distance $2d$, $d \geq 2$, from one with distance 2. Both codes will have the same number of codewords, the same maximum runlength and accumulated charge, and the number of checkbits blocks in the C -part will be at most quadratic in d . The computation of the checkbits requires some functions that are applied on the I -part Y . For a function f , the result of $f(Y)$ is either a residue modulo k or an element in some Galois field. Let $p_0 p_1 \cdots p_{2r-1} = \nu(f(Y))$ be the checkbits block P defined by the following.

- 1) If $f(Y)$ is a residue modulo k then $p_0 p_2 \cdots p_{2r-2}$ is the binary representation of $f(Y)$.
- 2) If $f(Y) \in GF(q)$ then let α be a primitive element in $GF(q)$. If $f(Y) = 0$ then $p_0 p_2 \cdots p_{2r-2}$ is the binary representation of $q-1$, and if $f(Y) = \alpha^i$, $0 \leq i \leq q-2$ then $p_0 p_2 \cdots p_{2r-2}$ is the binary representation of i .
- 3) $p_{2i+1} = \bar{p}_{2i}$ for $i = 0, \dots, r-1$, where $\bar{x}$ denotes the binary complement of x .

Note, that by this definition either $r = \lceil \log k \rceil$ or $r = \lceil \log q \rceil$. Also, ν^{-1} denotes the inverse function of ν . We give three different definitions for constructions of checkbits. Each definition is applied to different distances. Note, that all the definitions, lemmas, theorems, and procedures that follow can be applied to $(2n, 2d, n)$ codes.

Definition 1: Let $X = [YP]$, $Y = B_1 B_2 \cdots B_k$, be a word, where $B_i = b_{i,0} b_{i,1} \cdots b_{i,m-1}$, $1 \leq i \leq k$, is an arbitrary m -tuple of weight w_i , and P is a $(2r)$ -tuple of checkbits. Let

$$f(Y) = \sum_{i=1}^k \sum_{b_{i,j}=1} j(\text{mod } m)$$

and define P as $\nu(f(Y))$.

Lemma 1: Let $X_1 = [Y_1 P_1]$ and $X_2 = [Y_2 P_2]$ be two words with the form of Definition 1. If Y_1 and Y_2 are different in exactly two bits of block i then $d(P_1, P_2) \geq 2$.

The proof is similar to the proof of Theorem 1 in Graham and Sloane [3].

Proof: Assume Y_1 and Y_2 differ in bits s and t of block i (say bit s in Y_1 is 1), then $f(Y_1) \equiv z + s(\text{mod } m)$ and $f(Y_2) \equiv z + t(\text{mod } m)$ for some z . Hence $f(Y_1) \neq f(Y_2)$ and $d(P_1, P_2) \geq 2$. $\square$

From Lemma 1 one can infer the following.

Theorem 1: If C is a code whose codewords have $(w_1, \dots, w_k, w_{k+1})$ configuration and the form of words in Definition 1, then C has distance ≥ 4 .

Procedure 1 describes the error detection and correction for a code in which the codewords have the form of Definition 1. The correctness of this procedure can be easily verified.

Procedure 1: Let $\vec{a}$ be the information vector, $c(\vec{a})$ the transmitted codeword, and $\hat{c}(\vec{a}) = [YP]$ the received codeword. Let

$(u_1, \dots, u_k, u_{k+1})$ be the configuration of $\hat{c}(\vec{a})$ and let

$$e = \sum_{i=1}^{k+1} |u_i - w_i|$$

(note, that the w_i can be found even if there are a few configurations in the code, choose the configuration for which $e \leq 1$, if none exist at least two errors occurred). We distinguish between three cases.

Case 1: $e = 0$. If $P = \nu(f(Y))$ then take $\hat{c}(\vec{a})$ as the transmitted codeword. If $P \neq \nu(f(Y))$ then at least two errors occurred.

Case 2: If $e = 1$ then at least one error occurred.

a) $u_{k+1} \neq w_{k+1}$. If $d(P, \nu(f(Y))) = 1$ then take $[Y\nu(f(Y))]$ as the transmitted codeword. If $d(P, \nu(f(Y))) > 1$ then at least three errors occurred.

b) If $u_t \neq w_t$ for some $t < k+1$ then assume an error occurred in bit s of B_t and solve

$$\sum_{i=1}^k \sum_{b_{i,j}=1} j + s \equiv \nu^{-1}(P) \pmod{m}, \quad \text{if } u_t = w_t - 1$$

or

$$\sum_{i=1}^k \sum_{b_{i,j}=1} j - s \equiv \nu^{-1}(P) \pmod{m}, \quad \text{if } u_t = w_t + 1.$$

If either $u_t = w_t - 1$ and $b_{t,s} = 1$ or $u_t = w_t + 1$ and $b_{t,s} = 0$ then at least three errors occurred. Otherwise, take $\hat{c}(\vec{a})$, after setting $b_{t,s} := \bar{b}_{t,s}$, as the transmitted codeword.

Case 3: If $e > 1$ then at least two errors occurred.

Note, that in Procedure 1 and in Procedures 2 and 3 that follow, we did not take into consideration errors that can be detected by some specific properties of $\hat{c}(\vec{a})$, e.g., the parameters c and l of dc-free codes.

This method for computing checkbits can be extended for distances $2d > 4$. For these distances our I -part has one block. For $2d = 6$ the checkbits are computed as follows.

Definition 2: Let $X = [BP_1P_2]$ be a word, where

$$B = b_0b_1 \cdots b_{2m-1}$$

is an arbitrary $2m$ -tuple of weight m , and P_i , $1 \leq i \leq 2$, is a $(2r)$ -tuple of checkbits. Let q be a prime power such that $q \geq 2m + 1$, and let α be a primitive element in $GF(q)$. Let

$$f(B) = \sum_{b_j=1} j \pmod{q-1}$$

$$g(B) = \sum_{b_j=1} \alpha^j,$$

and define P_1 as $\nu(f(B))$ and P_2 as $\nu(g(B))$.

Lemma 2: Let $X_1 = [B_1P_1P_2]$ and $X_2 = [B_2Q_1Q_2]$ be two words formed by Definition 2.

- 1) If $d(B_1, B_2) = 2$ then $d(P_1, Q_1) \geq 2$ and $d(P_2, Q_2) \geq 2$.
- 2) If $d(B_1, B_2) = 4$ then $d(P_1P_2, Q_1Q_2) \geq 2$.

Proof:

- 1) Similarly to Lemma 1 it follows that if $d(B_1, B_2) = 2$ then $d(P_1, Q_1) \geq 2$ and $d(P_2, Q_2) \geq 2$.
- 2) Since $d(B_1, B_2) = 4$, there are four distinct coordinates t_1, t_2, s_1, s_2 such that

$$\begin{array}{cccccc} & t_1 & t_2 & s_1 & s_2 & \\ B_1 = \cdots & 1 & 1 & 0 & 0 & \cdots \\ B_2 = \cdots & 0 & 0 & 1 & 1 & \cdots \end{array}$$

and B_1 and B_2 agree in all other coordinates. If $P_1 = Q_1$ and $P_2 = Q_2$ then by Definition 2

$$t_1 + t_2 \equiv s_1 + s_2 \pmod{q-1}$$

$$\alpha^{t_1} + \alpha^{t_2} = \alpha^{s_1} + \alpha^{s_2}$$

which is possible if and only if $t_1 = s_1$ or $t_1 = s_2$ (the proof is omitted, it is similar to proofs in Golomb [13] and Etzion [14]), a contradiction. Hence, $d(P_1P_2, Q_1Q_2) \geq 2$. $\square$

From Lemma 2 one infers the following theorem.

Theorem 2: If C is a code whose codewords have the form $[BP_1P_2]$ and (w_1, w_2, w_3) configuration, where P_1 and P_2 were computed by Definition 2, then C has distance ≥ 6 .

Procedure 2 describes the error detection and correction for a code whose codewords have the form of Definition 2. The correctness of this procedure can be easily verified.

Procedure 2: Let $\vec{a}$ be the information vector, $c(\vec{a})$ the transmitted codeword, and $\hat{c}(\vec{a}) = [BP_1P_2]$ the received codeword. Let (u_1, u_2, u_3) be the configuration of $\hat{c}(\vec{a})$ and let

$$e = \sum_{i=1}^3 |u_i - w_i|.$$

We distinguish among four cases.

Case 1: $e = 0$.

- a) If $d(P_1, \nu(f(B))) + d(P_2, \nu(g(B))) \leq 2$ then take $[B\nu(f(B))\nu(g(B))]$ as the transmitted codeword.
- b) Assume two errors occurred in B , b_t is a 1-error and b_s is a 0-error. Solve

$$\begin{aligned} \sum_{b_j=1} j - t + s &\equiv \nu^{-1}(P_1) \pmod{q-1} \\ \sum_{b_j=1} \alpha^j - \alpha^t + \alpha^s &\equiv \nu^{-1}(P_2) \end{aligned}$$

or equivalently

$$\begin{aligned} s - t &\equiv k \pmod{q-1} \\ \alpha^s - \alpha^t &\equiv \beta \end{aligned}$$

where

$$k \equiv \nu^{-1}(P_1) - \sum_{b_j=1} j \pmod{q-1}$$

and

$$\beta \equiv \nu^{-1}(P_2) - \sum_{b_j=1} \alpha^j.$$

If $k = 0$, or $\beta = 0$, or $b_t = 0$, or $b_s = 1$, then at least four errors occurred. Otherwise, take $\hat{c}(\vec{a})$, after setting $b_t := 0$ and $b_s := 1$, as the transmitted codeword.

Case 2: If $e = 1$ then at least one error occurred.

- a) $u_2 \neq w_2$. If $P_2 \neq \nu(g(B))$ then at least three errors occurred. Otherwise, if $d(P_1, \nu(f(B))) > 1$ then at least three errors occurred. If $d(P_1, \nu(f(B))) = 1$ then take $[B\nu(f(B))\nu(g(B))]$ as the transmitted codeword.
- b) $u_3 \neq w_3$. This case is similar to (2a).
- c) $u_1 \neq w_1$. Assume an error occurred in b_s , and solve

$$\sum_{b_j=1} j + s \equiv \nu^{-1}(P_1) \pmod{q-1}, \quad \text{if } u_1 = w_1 - 1$$

or

$$\sum_{b_j=1} j - s \equiv \nu^{-1}(P_1) \pmod{q-1}, \quad \text{if } u_1 = w_1 + 1.$$

If either $u_1 = w_1 - 1$ and $b_s = 1$ or $u_1 = w_1 + 1$ and $b_s = 0$ then at least three errors occurred. Otherwise, set $b_s := \bar{b}_s$ and compute

$\nu(g(B))$. If $P_2 \neq \nu(g(B))$ then at least three errors occurred. Otherwise, take $\hat{c}(\vec{a})$, after the setting $b_s := \bar{b}_s$, as the transmitted codeword.

Case 3: If $e = 2$ then at least two errors occurred.

- $\sum_{i=2}^3 |u_i - w_i| = 2$. If $d(P_1, \nu(f(B))) + d(P_2, \nu(g(B))) = 2$ then take $[B\nu(f(B))\nu(g(B))]$ as the transmitted codeword. Otherwise, at least four errors occurred.
- $u_1 - w_1 = 2$. Assume two 1-errors occurred in b_l and b_s , solve

$$\sum_{b_i=0} i - t - s \equiv \nu^{-1}(P_1) \pmod{q-1}$$

$$\sum_{b_i=0} \alpha^i - \alpha^t - \alpha^s = \nu^{-1}(P_2),$$

and continue as in (1b).

- $u_1 - w_1 = -2$. This case is similar to (3b).
- $|u_1 - w_1| = 1$. This case is similar to (2c). Assume one error occurred in B and one error occurred in either P_1 or P_2 .

Case 4: If $e > 2$ then at least three errors occurred.

For distances $2d > 6$ the checkbits are defined as follows.

Definition 3: Let $X = [BP_1 \cdots P_{d-1}]$ be a word, where $B = b_0b_1 \cdots b_{2m-1}$ is an arbitrary $2m$ -tuple of weight m , and P_i , $1 \leq i \leq d-1$, is a $(2r \cdot (2d-1))$ -tuple of checkbits. Let s be the smallest integer such that $2^s \geq 2m$, and let α be a primitive element in $GF(2^s)$. Let

$$f_i(B) = \sum_{b_j=1} (\alpha^j)^{2^{i-1}}, \quad 1 \leq i \leq d-1$$

and define P_i as concatenation of $2d-1$ identical blocks with the value $\nu(f_i(B))$.

Lemma 3: Let $X_1 = [B_1P_1 \cdots P_{d-1}]$ and

$$X_2 = [B_2Q_1 \cdots Q_{d-1}]$$

be two words with the form of Definition 3. If $d(B_1, B_2) = 2k$ then $d(X_1, X_2) \geq 2d$.

Proof: Similar to the BCH bound [15] one can show that if $d(B_1, B_2) = 2k$, $2k < 2d$ then $[\nu(f_1(B_1)) \cdots \nu(f_k(B_1))]$ and $[\nu(f_1(B_2)) \cdots \nu(f_k(B_2))]$ are different in at least two bits. Since $P_i(Q_i)$ is a concatenation of $2d-1$ blocks with the value $\nu(f_i(B_1))\nu(f_i(B_2))$ it follows that $d(X_1, X_2) \geq 2d$. $\square$

An immediate result from Lemma 3 is the following theorem.

Theorem 3: If C is a code whose codewords have the form $[BP_1 \cdots P_{d-1}]$ and $(w_1, \dots, w_d)$ configuration, where the P_i 's were computed by Definition 3, then C has distance $\geq 2d$.

Procedure 3 describes the error detection and correction for a code whose codewords have the form of Definition 3. The correctness of this procedure can be easily verified.

Procedure 3: Let $\vec{a}$ be the information vector, $c(\vec{a})$ the transmitted codeword, and $\hat{c}(\vec{a}) = [BP_1 \cdots P_{d-1}]$ the received codeword. First, each P_i is divided into $2d-1$ blocks $P_i = [p_{i,1} \cdots p_{i,2d-1}]$, where $p_{i,j}$ is of length $2r$. For given i , let Q_i be a pattern among the $p_{i,j}$'s which appears at least d times. If none exist or $W(Q_i) \neq r$ then at least d errors occurred. Let $u = W(B)$ and $e = |u - w_1|$. We distinguish between two cases.

Case 1: If $e > d-1$ then at least d errors occurred.

Case 2: If $0 \leq e < d$ then for each l , $0 \leq l$ and $0 < e+2l < d$, assume $e+2l$ errors occurred and solve the following $e+2l$

equations

$$\nu^{-1}(Q_i) = \sum_{b_j=1} (\alpha^j)^{2^{i-1}} + \sum_{j=1}^{e+2l} (x_j)^{2^{i-1}}, \quad 1 \leq i \leq e+2l.$$

Let $x_j = \alpha^{k_j}$, $1 \leq j \leq e+2l$. If

$$w_1 = u - \sum_{j=1}^{e+2l} b_{k_j} + \sum_{j=1}^{e+2l} (1 - b_{k_j}) \quad (1)$$

then take $\hat{c}(\vec{a})$ as the transmitted codeword, after setting $b_{k_j} := \bar{b}_{k_j}$ for each j , $1 \leq j \leq e+2l$. If for each l , the $e+2l$ equations have no solution that satisfies (1) then at least d errors occurred.

The decoding of Procedures 1 and 2 is very simple, while solving the equations in Procedure 3 is done as in the decoding for BCH codes [15]. Also note, that the computation of the C-part from the I-part is very simple in all the procedures.

Now, let C be a $(2n, l, c, 2d)$ dc-free code. To each checkbits block of length $2r$ in C previously defined there are z possible combinations (m for Definition 1, $q-1$ or q for Definitions 2 and 3). Let s be an integer such that $s \leq r$ and $A(2s, l, c, 2) \geq z \cdot k$. If the related $(2s, l, c, 2)$ code can be partitioned into z classes with at least k words at mutual distance $\geq 2d$, we can use, similar to van Tilborg and Blaum [12], a $1 \rightarrow k$ mapping from the z combinations of checkbits (of one checkbits block) onto those $z \cdot k$ codewords of the $(2s, l, c, 2)$ code, i.e., each combination of checkbits is mapped onto a different class with k codewords. By applying this mapping to one checkbits block, the code will multiply its cardinality by k , while its length is reduced by $2r - 2s$. Those partitions can be constructed in similar methods to the ones in [3] and [7].

Example 1: Assume the codewords of a code C have the form $[B_1 \cdots B_k P]$, where B_i , $1 \leq i \leq k$, is of length 6, P was computed by Definition 1, $c = 2$, and $l = 4$. Since $A(4, 4, 2, 2) = 6$ we can map each one of the six combinations of checkbits (010101, 010110, 011001, 011010, 100101, 100110 for 0, 1, 2, 3, 4, 5, modulo 6) onto a different word of length four and weight 2 (0011, 0101, 0110, 1001, 1010, 1100). The length of the code is reduced by 2 to $6k + 4$, while its cardinality remains the same. Since $A(6, 4, 2, 2) = 18$ we can also use the following partition, of the code which attains $A(6, 4, 2, 2)$, into six classes with mutual distance ≥ 4 :

011100	011010	010110	001110	001101	001011
101010	100110	011001	010101	010011	101100
110001	101001	100101	100011	110100	110010

Each one of the six combinations of checkbits is mapped into a different class. Thus, the code multiply its cardinality by three, while its length remains $6(k+1)$. Note, that those 18 words can be partitioned into five classes with mutual distance ≥ 4 :

110010	110100	011010	100011	010101
011100	101001	100110	011001	101010
001011	010011	110001	101100	
100101	001110	001101	010110	

Lemma 4: Let C be a $(2n, l, c, 2d)$ dc-free code with checkbits obtained by Definition 1, 2, or 3. The code obtained by using the $1 \rightarrow k$ mapping also has distance $2d$.

Proof: Let $X_1 = [Y_1 P_1]$ and $X_2 = [Y_2 P_2]$ be two codewords in C , where $Y_1(Y_2)$ is a codeword excluding one checkbits block $P_1(P_2)$, on which the $1 \rightarrow k$ mapping is applied. We distinguish between two cases.

- a) $Z_1 = [Y_1 Q_1]$ and $Z_2 = [Y_1 Q_2]$ are two codewords obtained from X_1 (Q_1 and Q_2 belong to the same class with mutual distance $\geq 2d$). Since $d(Q_1, Q_2) \geq 2d$ it follows that $d(Z_1, Z_2) \geq 2d$.
- b) $Z_1 = [Y_1 Q_1]$ and $Z_3 = [Y_2 Q_3]$ are two codewords obtained from X_1 and X_2 , respectively. By Definitions 1, 2, and 3, if $d(Y_1, Y_2) < 2d$ then $d(Y_1, Y_2) = 2d - 2$ and $d(P_1, P_2) \geq 2$. Since Q_1 and Q_3 have the same weight and $Q_1 \neq Q_3$ it follows that $d(Q_1, Q_3) \geq 2$ and therefore $d(Z_1, Z_3) \geq 2d$. $\square$

III. DC-FREE CODES WITH ASYMPTOTIC RATES 0.7925 AND 0.8858

In this section we construct two dc-free codes. The first code is a $(2n, 4, 2, 2d)$ code, for a given d , $d \geq 1$, and its asymptotic rate is 0.7925. The second code is a $(2n, 6, 3, 2d)$, for a given d , $d \geq 1$, and its asymptotic rate is 0.8858. These rates achieve the channel capacity asymptotically as the code length goes to infinity [16].

Let $C_2(2n)$ denote all the words of a $(2n, 4, 2, 2)$ dc-free code that attains $F(2n) \triangleq A(2n, 4, 2, 2)$. Even length prefixes to a codeword in $C_2(2n)$ are of two types.

Type 1) The number of 0's and 1's is the same.

Type 2) The number of appearances of one symbol is greater by 2 than the number of appearances of the second symbol.

If a codeword begins with 01 or 10 then its suffix is a codeword from $C_2(2n-2)$. If a codeword begins with 00 (Type 2)) then its prefix is either 0011 or 0010. If the prefix is 0011 then its suffix is a codeword from $C_2(2n-4)$. If the prefix is 0010 then we have again a prefix of Type 2). If the prefix of the codeword is 11 the complement of this codeword is a codeword whose prefix is 00. If the prefix of the codeword is of Type 2) and has length $2n-2$, then the suffix of the codeword is determined uniquely. Therefore we have

$$F(2n) = 2F(2n-2) + 2(F(2n-4) + F(2n-6) + \dots + F(2) + 1) \quad (2)$$

or equivalently

$$F(2n) = 3F(2n-2) \quad (3)$$

with the initial condition $F(2) = 2$. The solution of the equation gives

$$F(2n) = 2 \cdot 3^{n-1}.$$

To each codeword of $C_2(2n)$ we will compute checkbits as described in Section II, to obtain distance $2d$, $d \geq 2$. Hence, the code will have length $2n + 2r(d-1)$ for $d \leq 3$ and length $2n + 2r(d-1)(2d-1)$ for $d > 3$, where $r \leq \lceil \log(2n+1) \rceil$, the maximum runlength is 4, and the maximum accumulated charge is 2. The rate of the code for $d \leq 3$ and $n \rightarrow \infty$ is

$$\frac{\log(2 \cdot 3^{n-1})}{2n + 2r(d-1)} > \frac{1 + (n-1)\log 3}{2n + 2(d-1)\lceil \log(2n+1) \rceil}.$$

Since $\log 3 = 1.585$ we have that asymptotically the rate is 0.7925. The same rate is obtained for $d > 3$ and $n \rightarrow \infty$.

Let $C_3(2n)$ denote all the words of a $(2n, 6, 3, 2)$ dc-free code that attains $G(2n) \triangleq A(2n, 6, 3, 2)$. Even length prefixes to a codeword in $C_3(2n)$ are of two types.

Type 1) The number of 0's and 1's is the same.

Type 2) The number of appearances of one symbol is greater by 2 than the number of appearances of the second symbol.

If a codeword begins with 01 or 10 then its suffix is a codeword from $C_3(2n-2)$. If a codeword begins with 00 (Type

2)) then its prefix is 0011, 0010, or 0001. If the prefix is 0011 then its suffix is a codeword from $C_3(2n-4)$. If the prefix is either 0010 or 0001 then we have again a prefix of Type 2). If the prefix of the codeword is 11 the complement of this codeword is a codeword whose prefix is 00. If the prefix of the codeword is of Type 2) and has length $2n-2$, then the suffix of the codeword is determined uniquely. Therefore we have

$$G(2n) = 2G(2n-2) + 2(G(2n-4) + 2(\dots 2(G(2) + 2))) \quad (4)$$

or equivalently

$$G(2n) = 4G(2n-2) - 2G(2n-4) \quad (5)$$

with the initial conditions $G(2) = 2$ and $G(4) = 6$. The solution of the equation gives

$$G(2n) = 0.5 \cdot (2 + \sqrt{2})^n + 0.5 \cdot (2 - \sqrt{2})^n.$$

To each codeword of $C_2(2n)$ we will compute checkbits as described in Section II, to obtain distance $2d$, $d \geq 2$. Hence, the code will have length $2n + 2r(d-1)$ for $d \leq 3$ and length $2n + 2r(d-1)(2d-1)$ for $d > 3$, where $r \leq \lceil \log(2n+1) \rceil$, the maximum runlength is 6, and the maximum accumulated charge is 3. The rate of the code for $d \leq 3$ and $n \rightarrow \infty$ is

$$\frac{\log(0.5 \cdot (2 + \sqrt{2})^n + 0.5 \cdot (2 - \sqrt{2})^n)}{2n + 2r(d-1)} > \frac{\log 0.5 + n \cdot \log(2 + \sqrt{2})}{2n + 2(d-1)\lceil \log(2n+1) \rceil}.$$

Since $\log(2 + \sqrt{2}) = 1.7716$ we have that asymptotically the rate is 0.8858. The same rate is obtained for $d > 3$ and $n \rightarrow \infty$.

For $l = 8$, $c = 4$, and $d = 2$, the recursion formula takes a much more complicated form. For $l = 2$, $c = 1$, and $d = 2$, each codeword has the form $[B_1 B_2 \dots B_n]$, where $B_i \in \{01, 10\}$. Hence to obtain a dc-free code with distance 4 we have to concatenate a block with only two check symbols (01 or 10). Thus we have a $(2n+2, n, 2, 1, 4)$ dc-free block code that has rate 0.5. Those codewords are also used in Blaum [9]. To obtain other distances we use Definitions 2 and 3, and obtain a $(2n+4r, 2, 1, 6)$ dc-free code and a $(2n+2r(d-1)(2d-1), 2, 1, 2d)$, $d > 3$, code with asymptotic rate 0.5. Finally, for $c = 1$ there is a strong connection between dc-free codes and (n, d) codes.

Theorem 4: $A(2n, 2, 1, 2d) = A(n, d)$.

Proof: From a given $(n, 4)$ code we construct a $(2n, 2, 1, 2d)$ code by using the substitution $0 \rightarrow 01$ and $1 \rightarrow 10$. Given a $(2n, 2, 1, 2d)$ dc-free code with codewords of the form $[B_1 \dots B_n]$, $B_i \in \{01, 10\}$, we construct an (n, d) code by using the substitution $01 \rightarrow 0$ and $10 \rightarrow 1$ on the B_i 's. Now, it can be easily verified that $A(2n, 2, 1, 2d) = A(n, d)$. $\square$

The recursion formulas (2) and (3) enable us to give simple procedures for the encoding and decoding of C_2 . Let $F(0) = 1$, and let $D(2n) = \sum_{i=0}^{n-2} F(2i)$. Let iv represent the integer value of the information word. For the encoding there are four cases.

Case 1: If the information word has value iv , such that $0 \leq iv < F(2n-2)$, then the first two bits of the codeword are 01. Continue with the value iv and the encoding related to $C_2(2n-2)$.

Case 2: If the information word has value iv , such that $F(2n-2) \leq iv < 2F(2n-2)$, then the first two bits of the codeword are 10 and set $iv := iv - F(2n-2)$. Continue with the value iv and the encoding related to $C_2(2n-2)$.

Case 3: If the information word has value iv , such that $2F(2n-2) \leq iv < 2F(2n-2) + D(2n)$, then the first two bits of the codeword are 00. Find the value of j , $j \geq 2$, such that, $2F(2n-2) + \sum_{i=2}^{j-1} F(2n-2i) \leq iv < 2F(2n-2) + \sum_{i=2}^j F(2n-2i)$. The next $2j-2$ bits are $j-2$ blocks of 10 followed by 11. Set $iv := iv - 2F(2n-2) - \sum_{i=2}^{j-1} F(2n-2i)$, and continue with the value iv and the encoding related to $C_2(2n-2j)$.

Case 4: If the information word has value iv , such that $2F(2n-2) + D(2n) \leq iv < 2F(2n-2) + 2D(2n)$, then the first two bits of the codeword are 11. Find the value of j , $j \geq 2$, such that, $2F(2n-2) + D(2n) + \sum_{i=2}^{j-1} F(2n-2i) \leq iv < 2F(2n-2) + D(2n) + \sum_{i=2}^j F(2n-2i)$. The next $2j-2$ bits are $j-2$ blocks of 01 followed by 00. Set $iv := iv - 2F(2n-2) - D(2n) - \sum_{i=2}^{j-1} F(2n-2i)$, and continue with the value iv and the encoding related to $C_2(2n-2j)$.

After the first $2n$ bits have been constructed, we compute the checkbits to obtain the desired distance by the appropriate definition of Section II. To shorten a block $P = p_0 p_1 \cdots p_{2r-1}$ of the checkbits, let s be the number of possible combinations for the checkbits, let t be the smallest value such that $F(2t) \geq s$, and let u be the integer value of $p_0 p_2 \cdots p_{2r-2}$. The new checkbits block Q is of length $2t$ and its value is word number u in $C_2(2t)$. Since a $1 \rightarrow 1$ mapping is used it follows by Lemma 4 that the distance of the code is unchanged.

Example 2: For $m=8$, $F(16)=4374$, and we want to find codeword number 3613. Since $2916 = 2F(14) \leq 3613 < 2F(14) + D(16) = 3645$, we have that $c_0 c_1 = 00$. Now, $3564 = 2F(14) + F(12) + F(10) \leq 3613 < 2F(14) + F(12) + F(10) + F(8) = 3618$. Hence, $c_2 c_3 c_4 c_5 c_6 c_7 = 101011$. We continue with $C_2(8)$ and $v = 3613 - 3564 = 49$. Since $45 = 2F(6) + D(8) \leq 49 < 2F(6) + 2D(8) = 54$, we have that $c_8 c_9 = 11$. Now $45 = 2F(6) + D(8) \leq 49 < 2F(6) + D(8) + F(4) = 51$. Hence, $c_{10} c_{11} = 00$. We continue with $C_2(4)$ and $v = 49 - 45 = 4$. Since $4 = 2F(2) \leq 4 < 2F(2) + D(4) = 5$, we have that $c_{12} c_{13} = 00$ and therefore $c_{14} c_{15} = 11$. Thus we obtain the codeword 0010101111000011. The computation of the checkbits by Definition 1 produces the bits 01010110. Now, since $F(6) \geq 16 > F(4)$ we can use only six checkbits. We have to find codeword number 1 in $C_2(6)$. This word is computed by the procedure of $C_2(6)$, and it is 010110. Thus codeword number 3613 is 0010101111000011010110.

The decoding is done first by correcting the errors using the appropriate procedure (we can detect more errors by using the fact that $c=2$), and then, using an inverse method to the encoding, by evaluating iv from the codeword.

Similarly, there is a procedure for the encoding of $C_3(2n)$ using (4) and (5).

IV. A (30, 21, 10, 6, 4) DC-FREE BLOCK CODE

The rates of the codes that were obtained in Section III achieve the channel capacity. But, for small lengths their rates are not the best. In the next two sections we consider dc-free codes that have good rates for small lengths. In Section V we will give a construction for length $4n$, $n \geq 4$. For length $4n+2$ we give in this section a construction for length 30.

We will consider each codeword as a concatenation of four blocks $[B_1 B_2 B_3 B_4]$, where B_1 , B_2 , and B_3 are 8-tuples each and B_4 is a 6-tuple. To achieve the desired parameters we will consider the following nineteen configurations: 1) (4, 4, 4, 3), 2) (2, 4, 6, 3), 3) (2, 6, 4, 3), 4) (4, 2, 6, 3), 5) (4, 6, 2, 3), 6) (6, 2, 4, 3), 7) (6, 4, 2, 3), 8) (3, 3, 5, 4), 9) (3, 5, 3, 4), 10) (5, 3, 3, 4), 11) (3, 5, 5, 2), 12) (5, 3, 5, 2), 13) (5, 5, 3, 2), 14) (4, 2, 4, 5), 15) (4, 6, 4, 1), 16) (4, 4, 2, 5), 17) (4, 4, 6, 1), 18) (2, 4, 4, 5), 19) (6, 4, 4, 1). Since for any

given two configurations (w_1, w_2, w_3, w_4) and (u_1, u_2, u_3, u_4) , $\sum_{i=1}^4 |w_i - u_i| \geq 4$, it follows that the distance between two codewords from two different configurations is at least 4.

In configurations 1 through 7, the bits of B_4 are the checkbits of the codeword. In configurations 8 through 13, the first six bits of the B_2 are the checkbits. In configurations 14 and 15 the first six bits of B_3 are the checkbits. In configurations 16 through 19 the first six bits of B_2 are the checkbits. The checkbits are computed by Definition 1. For this computation, if the six checkbits are from a block of length eight, then the other two bits, of this block, are appended at the end of B_4 and form a new block of length 8. By Lemma 1 the distance between two codewords from the same configuration is 4, and hence the distance of the code will be 4.

In configuration 1 there are $\binom{8}{4}^3 = 343\,000$ codewords. In each of the configurations 2–7 there are $\binom{8}{4} \binom{8}{2}^2 = 54\,880$ codewords. In each of the configurations 8–13 there are $\binom{8}{3} \binom{6}{2} = 47\,040$ codewords. In each of the configurations 14–19 there are $2 \binom{8}{4} \binom{8}{2} = 23\,520$ codewords. Thus, we have a total of 1\,095\,640 codewords.

The maximum runlength occurs when a codeword that ends with five 0's(1's) is followed by a codeword that begins with six 0's(1's). The maximum accumulated charge is 8, and it occurs in codewords from configuration 2, 7, 18, and 19. Thus we obtained a (30, 21, 8, 4) dc-free block code.

To obtain a code with a better maximum runlength and a better maximum accumulated charge we remove from the code all those codewords that cause maximum runlength 11 and all the codewords with maximum accumulated charge 7 or 8. First we remove all the codewords that end with a run of five. Those codewords are from configurations 14–19 in which the last five symbols in the last block form a run of length 5. There are $2 \binom{8}{4} \binom{8}{2} = 3\,920$ codewords of this form in each configuration and a total of 23\,520 codewords. Codewords with maximum accumulated charge of 7 or 8, have either the prefix $Yxxx$, where $x \in \{0, 1\}$, and the difference between the number of x 's and $\bar{x}$'s in Y is 4, or the prefix YZ where in Z there are four x 's and one $\bar{x}$, and the difference between the number of x 's and $\bar{x}$'s in Y is 4. In each of the configurations 2 and 7 there are $8 \binom{8}{2} = 6\,272$ codewords of this form. In each of the configurations 8 and 13 there are $\binom{8}{3} \binom{6}{2} = 840$ codewords of this form. In each of the configurations 18 and 19 there are $8 \binom{8}{2} \cdot 5 = 2\,160$ codewords of this form (after the previous 3\,920 codewords were removed). A total of 105\,356 codewords now remain. Now the maximum runlength is 10 and the maximum accumulated charge is 6. Another 5\,000 codewords are removed from configuration 19 and the 104\,856 codewords which remain form a (30, 20, 10, 6, 4) dc-free block code.

Now we double the cardinality of the code by considering a 1-2 mapping from the 8 possible combinations of the checkbits onto 16 out of 20 words of length 6 and weight 3 (000111, 111000, 100011, and 011100 are not used). Each combination of checkbits will be mapped onto two complementary words of length 6 and weight 3. It can be verified that the other parameters of the code do not change. Thus we obtain a (30, 21, 10, 6, 4) dc-free block code.

For the encoding and decoding of the (30, 21, 10, 6, 4) dc-free block code we will use some look-up tables and an enumeration

method that associates to each integer between 0 and $2^{21} - 1$ an appropriate codeword. Since the procedures are very specific they are omitted.

V. CODES OF LENGTH $4n$

In this section we consider $(n, 4)$ codes, i.e., codewords of length n , even weight, and distance 4. For information on codes of this form the reader is referred to [8], [17], and [18]. From a given $(n, 4)$ code we will construct a dc-free block code of length $4n$, maximum runlength 4, 5, or 6, maximum accumulated charge not more than $n + 1$ (but usually significantly better), and distance 4. If the $(n, 4)$ code has a simple construction (like the Hamming code [19]) then our dc-free code will have a relatively easy procedures for encoding and decoding. Each candidate word for the dc-free block code, is considered as a concatenation of n blocks $[B_1 B_2 \cdots B_n]$, where each block is a 4-tuple. Given the $(n, 4)$ code C , each codeword $c_1 c_2 \cdots c_n \in C$ will correspond to a few configurations of the dc-free block code. If $c_i = 0$ then $w_i = 2$ for all these configurations, and if $c_i = 1$, then w_i will be either 1 or 3. Since the weight of a codeword is half of its length, the number of blocks with weight 1 is equal to the number of blocks with weight 3. Clearly the distance between two candidate words for the dc-free code, from two different configurations, is at least 4. For the distance between two words from the same configuration to be at least 4, each codeword must have one block of checkbits. For the configurations in which all the blocks have even weight, the bits of the last block are the checkbits. They are computed as in van Tilborg and Blaum [12], and hence we can obtain $2 \cdot 6^{n-1}$ words from these configurations. For the other configurations, the bits of the last block of odd weight in each configuration are the checkbits, and they are computed by Definition 1, and 1 1-1 mapping from the four possible combinations of checkbits {0101, 1010, 0110, 1001} onto the four possible blocks of length 4 and either weight 1 or weight 3. The maximum runlength of a symbol can arise from a concatenation of two blocks with the same odd weight, and hence it will be 6. The maximum accumulated charge can arise from a sequence of blocks with the same odd weight $w = 1(3)$, followed by a block that begins with either two or three 0's(1's). Hence, it can be easily verified that the maximum accumulated charge can be $n + 1$. To obtain a $(4n, k, l, c, 4)$ code we have to remove a few codewords. We remove codewords that cause runlengths 6 (and if possible also 5), to obtain maximum runlength 5 (or 4), and codewords with the highest accumulated charge.

Example 3: For $n = 4$, {0000, 1111} is a $(4, 4)$ code. From the codeword 0000 we obtain $2 \cdot 6^3$ codewords of length 16, with $c = 2$ and $l = 4$, by the method of van Tilborg and Blaum [12]. For the codeword 1111 we consider configurations (1, 3, 1, 3), (3, 1, 3, 1), and (1, 3, 3, 1). From each configuration we can take 27 codewords to form a dc-free block code with $c = 2$ and $l = 4$. We take 80 out of these 81 codewords to obtain a $(16, 9, 4, 2, 4)$ dc-free code. This is an improvement on the $(16, 9, 6, 5, 4)$ dc-free code introduced by Blaum [10].

Similar to Example 3 we obtain a code of length 20 and $k = 12$ from the $(5, 4)$ code {00000, 01111}. For $4n = 24$ we obtain a code with $k = 16$ from the $(6, 4)$ code {000000, 001111, 110011, 111100}. For $4n = 28$ we obtain a code with $k = 19$ from the $(7, 4)$ code {0000000, 0001111, 1010101, 1101001, 0110011, 0111100, 1100110, 1011010}. For $4n = 32$ we obtain a code with $k = 22$ from the $(8, 4)$ Hamming code [19]. For $4n = 64$ we are using the $(16, 4)$ Hamming code, and we

TABLE I
BOUNDS ON $\lfloor \log A(2n, 4, n) \rfloor$ AND k OF OUR DC-FREE CODES

$2n$	Upper Bound [20]	Best Known Lower Bound [3], [4], [7], [8]	k of Our DC-Free Code
16	10	10	9
20	14	13	12
24	17	17	16
28	21	20	19
30	23	22	21
32	25	24	22
64	55	54	51

form a dc-free code with $k = 51$. The values of maximum runlength and accumulated charge are determined in a way that the code will have relatively easy procedures for encoding and decoding. All the results that we have obtained, together with the lower and upper bounds are summarized in Table I. The encoding and decoding procedures for the codes are very specific and hence left for the reader.

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